LLARS: Enabling Domain Experts & Developers Collaboration for LLM Prompting, Generation and Evaluation

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Abstract

We demonstrate LLARS, an open-source platform that bridges the gap between domain experts and developers for building LLM-based assistants. LLARS integrates three tightly connected modules into an end-to-end pipeline: **Collaborative Prompt Engineering** for real-time co-authoring with version control and instant LLM testing, **Batch Generation** for configurable output production across user-selected prompts × models × data with cost control and **Evaluation** for human and LLM evaluators to rate, rank, label or compare outputs with live inter-rater agreement metrics. New prompts and models are automatically available for batch generation and completed batches can be turned into evaluation scenarios with a single click. Interviews with six domain experts and three developers in online counselling confirmed that LLARS eliminates the "translation work" between disciplines and enables domain experts to directly co-author prompts and conduct evaluations independently.

Demo: https://llars-project.github.io/demo

1 Introduction

Organizations building LLM-based assistants for sensitive domains [Chen *et al.*, 2024; Ling *et al.*, 2025]—online counselling [Stade *et al.*, 2024], legal analysis [Katz *et al.*, 2024], medical documentation [Van Veen *et al.*, 2024]—must write effective prompts, select suitable models and rigorously evaluate outputs, requiring both domain expertise and technical skill [Zamfirescu-Pereira *et al.*, 2023; Schulhoff *et al.*, 2024]. Public benchmarks rarely transfer to domain-specific tasks; practitioners must test models on their own data to find the smallest—and most cost-effective—option that still meets quality requirements, especially when replacing a model in a running system. Yet domain experts and developers typically work in isolation: prompts are iterated in shared documents without version control and no single tool covers the full cycle from prompt development through systematic evaluation.

We present LLARS (LLM Assisted Research System), an open-source platform that closes this gap with three tightly connected modules (Figure 1). *Collaborative Prompt Engineering* lets domain experts and developers co-author prompts in a shared real-time editor with version control and instant LLM testing. *Batch Generation* produces the user-configured Cartesian product of prompts × models × data items with cost estimation and budget control. *Evaluation* lets human and LLM evaluators rate, rank, label or compare outputs with live agreement metrics. New prompts and models automatically appear in batch generation and completed batches become evaluation scenarios with a single click; each module also supports standalone JSON/CSV export and connects to any OpenAI-compatible provider.

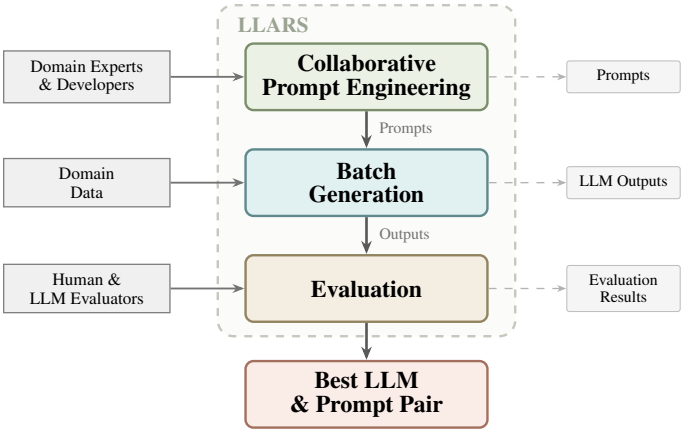

Figure 1: LLARS pipeline: domain experts and developers collaboratively develop prompts, generate outputs across LLMs and evaluate results with human and LLM evaluators. Each stage supports export; the pipeline yields a validated model–prompt combination.

Table 1 surveys twelve existing platforms along these dimensions. Agenta [Agenta AI, 2024] combines prompt versioning with LLM-as-judge scoring and provides A/B deployment testing, but lacks real-time co-editing and structured human evaluation with inter-rater agreement metrics. Phoenix [Arize AI, 2024], Langfuse [Langfuse GmbH, 2024] and W&B Weave [Weights & Biases, 2024] provide observability dashboards with dataset experiments and per-output feedback but focus on tracing and debugging production systems rather than systematic batch generation or coordinated multi-evaluator campaigns. ChainForge [Arawjo *et al.*, 2024]

Table 1: Comparison of GUI-first tools for prompt engineering, batch generation and hybrid LLM output evaluation. ● = native support; ◐ = partial; ○ = not supported. **Collab.** = real-time collaborative prompt co-editing; **Batch** = systematic generation over prompts × models × data; **H-Eval** = structured human evaluation campaigns; **LLM-E** = LLM-as-evaluator; **IRR** = inter-rater reliability metrics; **E2E** = integrated end-to-end pipeline.

Tool	OSS	Collab.	Batch	H-Eval	LLM-E	IRR	E2E
LLARS	●	●	●	●	●	●	●
Agenta [Agenta AI, 2024]	●	◐	◐	◐	●	○	◐
ChainForge [Arawjo *et al.*, 2024]	●	○	●	○	◐	○	◐
Phoenix [Arize AI, 2024]	●	○	◐	◐	●	○	○
Langfuse [Langfuse GmbH, 2024]	●	○	◐	◐	●	◐	○
W&B Weave [Weights & Biases, 2024]	●	○	◐	◐	●	○	○
Label Studio [HumanSignal, Inc., 2024]	●	○	○	●	◐	◐	○
Argilla [Argilla, Inc., 2024]	●	○	○	●	◐	◐	○
LangSmith [LangChain, Inc., 2024]	○	○	◐	◐	●	○	◐
Braintrust [Braintrust Data, Inc., 2024]	○	○	◐	◐	●	○	◐
Maxim [Maxim AI, 2024]	○	○	◐	◐	●	○	◐
Vellum [Vellum AI, 2024]	○	○	◐	◐	●	○	◐

offers a visual node-based interface for multi-model comparison with template variables and supports programmatic evaluation via custom code nodes, but omits structured human evaluation entirely. Label Studio [HumanSignal, Inc., 2024] and Argilla [Argilla, Inc., 2024] offer dedicated human annotation workflows—Label Studio computes Krippendorff's α across annotators, Argilla provides agreement analytics via its API—but both require externally generated outputs, disconnecting evaluation from prompt development. LangSmith [LangChain, Inc., 2024], Braintrust [Braintrust Data, Inc., 2024], Maxim [Maxim AI, 2024] and Vellum [Vellum AI, 2024] provide playground-style testing with dataset-level scoring but are closed-source, lack collaborative editing and omit batch generation. LLARS is an open-source platform to unify collaborative prompt engineering, Cartesian-product batch generation and hybrid human–LLM evaluation with agreement analytics, enabling developers and domain experts to work together from prompt development through rigorous assessment.

2 Platform Overview

Co-designed with developers, social science researchers and practising counsellors, LLARS runs as a containerized web application for textual data. Teams register proprietary or self-hosted OpenAI-compatible LLM providers; registered models then appear across all three modules.

2.1 Collaborative Prompt Engineering

The prompt engineering module (Figure 2) lets domain experts and developers co-author prompts in a real-time collaborative editor built on Yjs [Jahns, 2023], where every keystroke is instantly visible to all connected users.

A prompt is composed of ordered blocks, each representing one part of the prompt such as a data format description or an output instruction. One block may be marked as a system block that becomes the LLM's system prompt, while all remaining blocks are concatenated into the user prompt. Every block maintains its own version history tracking insertions

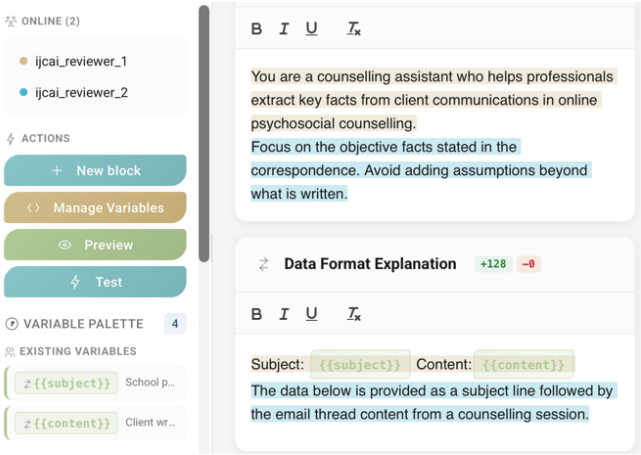

Figure 2: Collaborative prompt editor with a data-format explanation block containing template variables {{`subject`}} and {{`content`}}. The sidebar (left) shows actions for block management, variable management, preview and LLM testing. The Variable Palette (bottom-left) lists all extracted variables with sample values that can be dragged into blocks.

and deletions per save (visible as `+130 / −0` in Figure 2), enabling diff comparison and rollback without affecting other blocks. Blocks may contain template variables written as {{`variable_name`}} that act as placeholders for external data—an entire email thread, for instance, can be captured in a single {{`content`}} variable, keeping the surrounding instructions concise. All variables are automatically collected in the shared Variable Palette (Figure 2, bottom-left), from which they can be dragged into any block and annotated with sample values (e.g. {{`subject`}} = *"School problems at home"*). These sample values are substituted when testing the prompt against a selected LLM, with responses streaming back in real time so that multiple team members can test different models in parallel. Prompts can also be downloaded as JSON for use outside LLARS, but within the platform they

are directly available in batch generation, where the template variables are filled from dataset columns to scale a single prompt to hundreds of outputs.

2.2 Batch Generation

While the prompt editor allows rapid testing with sample values, a single example cannot validate whether a prompt generalizes across diverse inputs. Batch generation addresses this: users select from available prompts—self-authored or shared by collaborators—and any registered model to apply to their data, producing the Cartesian product of prompts $\times$ models $\times$ data items. A generation matrix previews this combination and estimates cost based on model pricing and projected token counts; optional budget caps pause jobs that exceed a threshold. A wizard offers three data source modes: upload data (CSV or JSON whose columns map to the prompt's template variables), previous generation (reuse data from a completed batch to test new prompts or models on the same dataset) and prompt only (generate without source data, filling variables from sample values). For example, 50 counselling email threads combined with two prompts and two models produce $50 \times 2 \times 2 = 200$ outputs.

Results stream in real time and every output records its full provenance—source item, prompt version, model, parameters, token count and cost—exportable as CSV or JSON. Evaluators see only the generated text without any indication of which model or prompt produced it; completed batches transfer into evaluation with a single click, preserving all attribution for downstream provenance analysis.

2.3 Evaluation

Each evaluation is organized as a *scenario*—a self-contained campaign that defines the type, assigns evaluators and distributes items—supporting six types: rating, ranking, labeling, comparison, mail rating and authenticity. Scenarios are created from batch outputs with a single click or via an AI-assisted Scenario Wizard; owners control whether all items go to every evaluator or configurable subsets are distributed for faster throughput. Where most tools limit LLMs to pairwise judging [Zheng *et al.*, 2024], LLARS treats them as full participants alongside humans.

The default ranking mode is *bucket ranking*: evaluators sort items into ordinal categories (e.g. *Good*, *Medium*, *Bad*) and rank within each bucket—reducing cognitive load [Miller, 1956] and yielding more reliable judgments than total ordering [Kiritchenko and Mohammad, 2017], as confirmed by domain expert interviews. Traditional ranking remains available.

Once assessments are submitted, scenario owners inspect aggregated results: live agreement metrics (Krippendorff's α, Cohen's and Fleiss' κ, Kendall's τ/W, ICC) filterable by evaluator pairing, bucket distribution charts and pairwise agreement heatmaps with click-through detail views. The central outcome is the *provenance analysis* (Figure 3): since batch generation records which model and prompt produced each output, ratings are traced back to their source and every combination is ranked by top-bucket hit rate—across three perspectives (per model, per prompt and combined)—directly answering the practitioner's question: *which pair should we deploy?* All data and metrics are exportable as JSON or CSV.

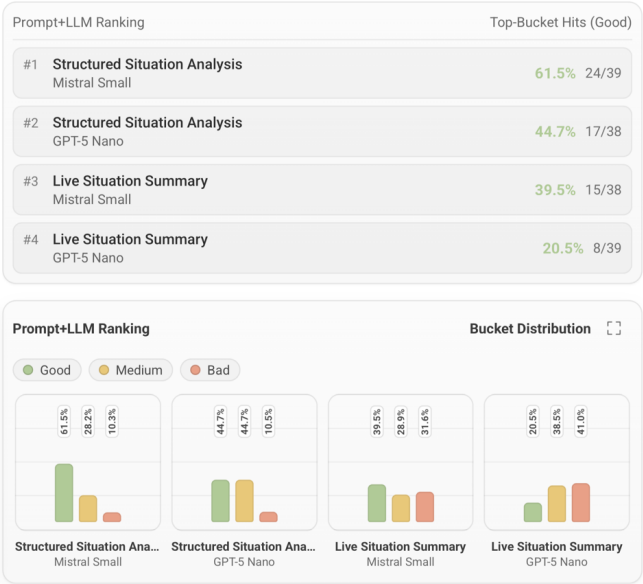

Figure 3: Provenance analysis ranking model–prompt combinations by top-bucket hit rate (top) with per-combination bucket distributions (bottom).

3 Deployment and User Study

LLARS is actively used in online counselling research: the Virtual Client project [Rudolph *et al.*, 2024] uses it for prompt development of simulated client interactions, and CAIA [Steigerwald *et al.*, 2025], an AI assistant for email counselling, relies on it for prompt development and evaluation. A full end-to-end case study [Steigerwald and Albrecht, 2025] produced 253 subject lines across 11 LLMs and had five professionals with an LLM evaluator rate them (1,518 assessments), determining how small a model could be while still meeting quality requirements.

Semi-structured interviews with six domain experts and three developers confirmed that consolidating the pipeline in one platform saves considerable time. Domain experts found prompt authoring straightforward and evaluation immediately understandable; developers valued that prompts, data and outputs remain in one place. Both groups emphasised that co-locating the workflow removed the "translation work" between disciplines.

4 Demo and Conclusion

Our demo walks attendees through the complete workflow—from collaborative prompt authoring through batch generation to evaluation with live agreement metrics, pairwise heatmaps and the provenance analysis shown in Figure 3—using pre-built counselling scenarios for hands-on exploration. By unifying prompt engineering, batch generation and evaluation, LLARS bridges the gap between domain experts and developers for interdisciplinary research.

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